SIGNDUPLEX: A FULL-DUPLEX INTERACTION CONTROLLER FOR REAL-TIME SIGN LANGUAGE COMMUNICATION

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ABSTRACT

Human–computer interaction requires not only understanding what users express but also identifying turn states in real time and handling new user input. Current intelligent sign language systems still lack a full-duplex interaction control mechanism for continuous visual input, and real-time interaction data remain scarce. We introduce SignDuplex, the first plug-and-play full-duplex interaction control framework for sign language. SignDuplex divides interaction into five events—REJECT, INCOMPLETE, WAIT, COMPLETE, and BARGE—and jointly models appearance and temporal motion through dual RGB–Pose visual encoding. To address the shortage of interaction data, we propose a visual–semantic data construction pipeline and build SignTurn, a dataset comprising 263,890 interaction event samples with a total duration of 57.6 h. Experiments show that SignDuplex achieves 94.99% overall accuracy in five-class interaction event recognition. In downstream sign language translation, SignDuplex yields a 67.3% relative improvement in BLEU-4 over whole-video offline translation on continuous multi-sentence input, while reducing first submission latency by 60.4%. Thus, independently modeling interaction timing improves translation quality and responsiveness for continuous signing, providing a new research direction for full-duplex human–computer interaction through sign language. The data and model will be made publicly available on GitHub.[1]

***Index Terms*—** Full-duplex sign language interaction, turn management, plug-and-play control, interaction data construction

1. INTRODUCTION

As multimodal large models advance, human–computer interaction is evolving from single-turn question answering toward continuous, real-time natural interaction. Recent full-duplex dialogue research addresses real-time control problems such as continuous listening, end-of-turn detection, and user interruption, whereas existing sign language research primarily focuses on content understanding. During continuous sign language interaction, a brief pause within an utterance may be mistaken for the end of a turn. The system must take over promptly after a complete utterance and detect new valid signing input initiated by the user during system output. A sign language agent must therefore understand the expressed content and continuously determine when to wait, when to take the turn, and when to respond to new user input.

To address these issues, we propose SignDuplex, a plug-and-play full-duplex interaction control framework for signing input. As shown in Fig. 1, SignDuplex decouples interaction timing decisions from downstream content generation, allowing the control module to focus on real-time interaction event prediction. Our main contributions are as follows:
(1) We are the first to formulate full-duplex control as a real-time interaction event prediction task for continuous signing, and introduce SignDuplex. Through dual RGB–Pose visual encoding, SignDuplex achieves 94.99% overall event recognition accuracy on the test set.
(2) We establish the first five-class interaction event definitions for full-duplex sign language interaction and integrate public American Sign Language (ASL) dialogue corpora, including MoLo, O5S5, and LING BING [1, 2, 3]. We also introduce sentence-level ASL translation data such as How2Sign [4] to supplement training samples, constructing the SignTurn dataset with 263,890 interaction event samples, including 260,400 training samples and 3,490 test samples, with a total duration of 57.6 h.
(3) We propose the first plug-and-play sign language interaction control design that decouples interaction decisions from downstream tasks. Experiments show that SignDuplex preserves sentence-level translation performance and improves translation quality and responsiveness for continuous multi-sentence input without modifying the downstream translation model.

2. RELATED WORK

2.1. Offline and Online Sign Language Translation

Sign language translation aims to convert continuous signing videos into natural language text. In recent years, research has gradually shifted from gloss supervision to gloss-free end-to-end modeling, using pretrained language models to enhance visual-to-language mapping. Sign2GPT introduces large language models for gloss-free translation [5], while FLa-LLM improves translation through visual representation learning and language model adaptation [6]. For real-time translation, SimulSLT combines wait-k, boundary prediction, and re-encoding to balance translation quality and latency [7]. However, its focus remains on generating translations in real time rather than identifying and controlling turn states in continuous interaction. Existing sign language translation models still exhibit substantial performance degradation on long sentences and continuous multi-sentence input [8], indicating that sentence-level translation methods cannot readily adapt to continuous interaction scenarios.

2.2. Full-Duplex Dialogue and Turn Management

Turn transitions in signed conversations are often accompanied by visual temporal cues such as utterance onsets and offsets, posture holds, and overlapping turns [9, 10]. In comparison, existing full-duplex interaction research mainly focuses on speech. One approach uses an independent turn-state detector to decouple interaction decisions from response content. Easy Turn combines acoustic and linguis-

[1]https://github.com/X-Agentic/SignDuplex

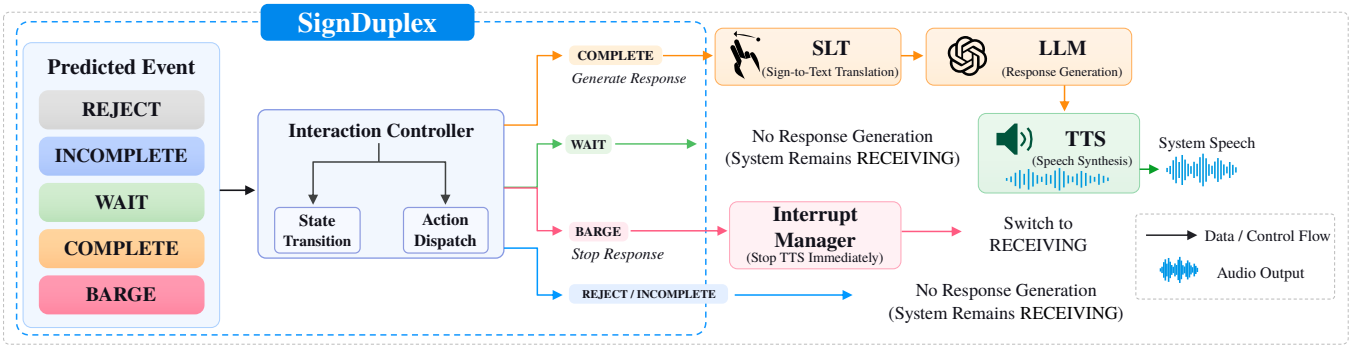


Fig. 1. SignDuplex interaction control.

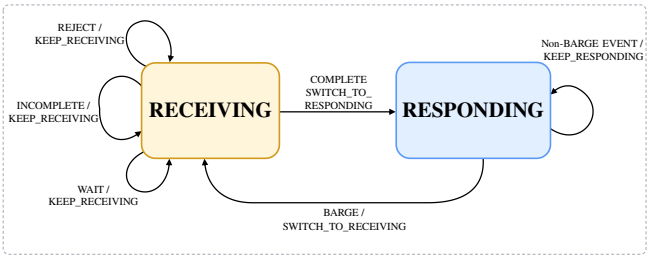


Fig. 2. Relationships between full-duplex system states and interaction events.

tic information for turn-state detection [11]; FireRedChat combines voice activity detection, semantic end-of-turn detection, and user interruption detection to achieve independent interaction control [12]. The second directly models full-duplex behavior in end-to-end models, such as Moshi [13], OmniFlatten [14], MiniCPM-o 4.5 [15], and Lychee-FD [16], but typically relies on large-scale full-duplex interaction data. Given the limited availability of such data for sign language, we adopt independent interaction control and formulate interaction timing decisions as a real-time prediction task.

3. PROPOSED METHOD

3.1. Interaction States and Event Routing

SignDuplex maintains two system states, RECEIVING and RESPONDING, representing the reception of user signing input and the execution of system output, respectively. Visual perception runs continuously in both states, ensuring that new user input can still be perceived during a response.

During online operation, SignDuplex makes streaming predictions at 1 s intervals. The model takes as input the RGB video accumulated since the most recent system state transition, the corresponding pose sequence, and the current system state. After a state transition, input accumulation restarts at the transition point. The state transitions are shown in Fig. 2.

In the RECEIVING state, REJECT, INCOMPLETE, and WAIT all maintain the current state. Detecting COMPLETE switches the system to RESPONDING and initiates the subsequent response. In the RESPONDING state, detecting BARGE immediately stops the current output and switches the system back to RECEIVING.

3.2. Dual RGB–Pose Modeling

Prior work shows that RGB and pose information are complementary for sign language understanding [17, 18, 19]. SignDuplex likewise retains both forms of visual information. For the same continuous RGB video stream, two visual pathways extract complementary information in parallel: the RGB branch primarily preserves fine-grained visual information such as facial expressions, changes in gaze, and local appearance, while the pose branch explicitly models the temporal motion of keypoints on the hands, face, and other regions. The two representations are converted into separate token sequences, concatenated with system state tokens, and passed to the language model for joint modeling. The overall model architecture is shown in Fig. 3.

The RGB branch uniformly samples the currently accumulated video and processes it through a SigLIP Vision Encoder [20] and Visual Resampler to obtain the RGB token sequence: $Z_k^R \in \mathbb{R}^{N_R \times D}$.

Here, Z_k^R is the RGB token sequence at streaming update k, N_R is the number of tokens, and D is the shared token feature dimension. The pose branch uses RTMPose [21] to extract hand and facial keypoint sequences from RGB video over the same time interval, and passes them through an ST-GCN Encoder [22], a Temporal Adapter, and a Projector to obtain pose tokens: $Z_k^P \in \mathbb{R}^{N_P \times D}$.

Here, Z_k^P is the pose token sequence encoded from the same accumulated video, and N_P denotes the number of tokens produced by the pose branch. The current system state is encoded as follows: $Z_k^S \in \mathbb{R}^{N_S \times D}$.

We set $N_S = 1$. The three types of tokens are concatenated along the sequence dimension:

$$Z_k = \text{Concat}(Z_k^R, Z_k^P, Z_k^S) \in \mathbb{R}^{(N_R+N_P+N_S)\times D}. \quad (1)$$

The concatenated sequence is passed to Qwen3-0.6B [23] to predict the current interaction event. Training uses a cross-entropy loss: $\mathcal{L}_{\text{cls}} = -\frac{1}{M}\sum_{i=1}^{M} \log p_i(y_i)$.

Here, M is the number of training samples, y_i is the ground-truth label, and $p_i(y_i)$ is the predicted probability of the corresponding class.

3.3. Interaction Event Definitions

Prior research shows that continuous hand movements can be divided into distinct motion phases. Kita et al. proposed a movement-phase coding method that provides a basis for analyzing the onset, continuation, holds, and termination of signing movements [24]. Computer-vision-based sign language analysis further uses keypoint motion and temporal changes to localize valid signing activity. Koulierakis et al. use keypoint information to identify intervals of actual signing in videos [25]. Moryossef et al. detect continuous signing activity from temporal changes in body pose [26]. Building on these studies, we define interaction events using hand motion velocity, expression progress, and local motion continuity:

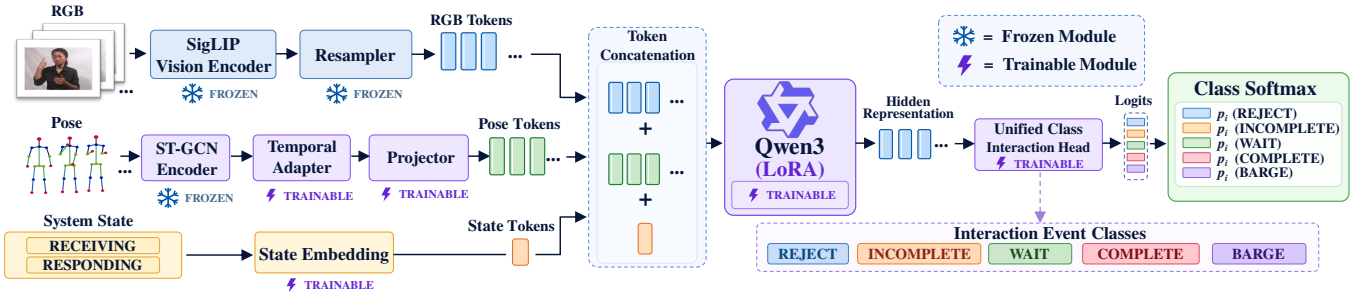


Fig. 3. Overview of the SignDuplex architecture.

1) **REJECT** indicates that the current hand movements do not constitute a valid signed utterance requiring a system response. This includes non-signing movements and brief backchannels that do not require the system to take the turn.
2) **INCOMPLETE** indicates that the user has begun a valid signed utterance and that the utterance is still in progress.
3) **WAIT** indicates brief deceleration, a posture hold, or a pause within an unfinished signed utterance.
4) **COMPLETE** indicates that the user has finished the current signed utterance and will not continue signing within this turn.
5) **BARGE** indicates that the user begins a new valid signed utterance while the system is producing output.

3.4. Sign Language Interaction Data Construction

Existing public sign language data mainly comprise real-world dialogue data and sentence-level translation data. For the former, we construct interaction event samples directly from existing annotations. For sentence-level data without turn information, we generate candidate events from hand motion features and verify them using semantic information.

To construct user signing videos as model input, we first designate the two signers in dialogue data with complete text and timestamp annotations as the user and the assistant, respectively. We then use SAM 2 [27] to segment and track videos showing both signers in the same frame, obtaining videos of the user alone while retaining the corresponding text annotations and the temporal relationships between the two signers' utterances. To construct labels for these inputs, we use DeepSeek-V3 [28] to identify instances of the five interaction events from the user's text for each input segment and generate event labels at the corresponding positions, yielding real-world dialogue interaction samples. Here, INCOMPLETE denotes ongoing signing whose meaning is not yet complete; WAIT denotes hesitation or a brief pause during an utterance; COMPLETE denotes the end of a complete utterance; BARGE denotes a new valid utterance initiated by the user before the assistant has finished; and REJECT denotes the absence of valid sign language semantics or a user backchannel.

For sentence-level data, we use RTMPose to extract 42 keypoints across both hands from RGB video. For a video containing N frames, let $p_{t,j}$ denote the normalized coordinates of hand keypoint j in frame t. The mean hand displacement between adjacent frames is defined as

$$x_t = \frac{1}{42} \sum_{j=1}^{42} \|p_{t,j} - p_{t-1,j}\|_2, \quad t = 2, \ldots, N. \tag{2}$$

Here, x_t denotes the mean magnitude of hand motion between adjacent frames. To distinguish low-motion states at different stages of an utterance, we further define cumulative motion progress:

$$\rho_t = \frac{\sum_{i=2}^{t} x_i}{\sum_{i=2}^{N} x_i}, \quad t = 2, \ldots, N. \tag{3}$$

Here, ρ_t is the ratio of the current cumulative motion to the total motion over the video.

We compute the distribution of x_t exclusively on the real-world dialogue training set and define its 25th and 75th percentiles as the low-motion threshold τ_{low} and active-motion threshold τ_{act}, respectively. A frame is marked active if $x_t \geq \tau_{\text{act}}$, and the active-frame ratio within a local interval $[a, b]$ is defined as $\alpha[a, b] = \frac{N_{\text{active}}[a,b]}{b-a+1}$.

Here, $N_{\text{active}}[a, b]$ is the number of active frames within the interval, and $\alpha[a, b]$ measures local motion continuity. The motion thresholds and candidate selection parameters above are all determined from statistics of the real-world dialogue training set and used to generate candidate events in sentence-level data. Based on these statistics, we generate candidate positions for different events along the timeline. INCOMPLETE candidates are generated when $\rho_t < 0.85$ and $\alpha[a, b] \geq 0.5$. When $\rho_t < 0.85$, $x_t \leq \tau_{\text{low}}$, and $\alpha[a, b] < 0.5$, low-motion intervals lasting 0.25–0.80 s are selected as WAIT candidates. COMPLETE candidates are generated when $\rho_t \geq 0.85$, $x_t \leq \tau_{\text{low}}$, and local motion continues to decline. For a candidate time t, we use only the video observed up to that time and obtain text using Uni-Sign [19]. DeepSeek-V3 then determines whether the semantic state agrees with the candidate label. Only samples satisfying both the motion and semantic conditions are retained.

Because sentence-level data contain relatively few WAIT and REJECT samples, we construct local slowing and brief-pause clips using video frame interpolation [29] and temporal resampling [30], and use Wan2.1 [31] to generate high-motion non-signing videos. The augmented samples are used only for training after motion and semantic verification. To ensure that the test set is independent of the automatic training data construction process, we use DeepSeek-V3 to assist in generating two-person dialogue scripts covering the five event classes, and two sign language users with five years of experience record continuous face-to-face conversations. The scripts specify the event types and interaction sequence in advance. Test labels are determined from the predefined events and their actual occurrence boundaries in the recordings.

4. EXPERIMENTS

4.1. Experimental Setup and Main Results

The model is trained on eight NVIDIA RTX 4090 GPUs for 8 epochs, with a batch size of 8. The SigLIP Vision Encoder, Resampler, and ST-GCN Encoder remain frozen; only the Temporal Adapter, Projector, State Embedding, classification head, and Qwen3-0.6B LoRA

Table 1. Interaction recognition and real-time performance.

Method	ACC_{rj} ↑	ACC_{incp} ↑	ACC_{wt} ↑	ACC_{cp} ↑	ACC_{bg} ↑	E2E ↓ (ms)
SignDuplex	96.88%	93.52%	93.06%	97.53%	96.00%	433.46

Table 2. SignTurn dataset size.

Event	Training set	Test set	Duration (h)
REJECT	46,820	640	9.8
INCOMPLETE	116,540	1,250	25.6
WAIT	41,760	720	9.6
COMPLETE	48,930	730	10.7
BARGE	6,350	150	1.9
Total	260,400	3,490	57.6

Table 3. RGB and pose modality ablation results.

Visual input	ACC_{rj} ↑	ACC_{incp} ↑	ACC_{wt} ↑	ACC_{cp} ↑	ACC_{bg} ↑
RGB-only	93.44%	89.04%	89.86%	91.10%	88.67%
Pose-only	92.34%	92.16%	87.92%	94.79%	93.33%
RGB+Pose	96.88%	93.52%	93.06%	97.53%	96.00%

parameters are updated. The Temporal Adapter, Projector, State Embedding, and classification head use a learning rate of 1×10^{-4}; LoRA uses a learning rate of 1×10^{-5}.

SignTurn contains 260,400 training samples and 3,490 manually annotated test samples; the distribution is shown in Table 2. Table 1 reports SignDuplex's interaction event recognition performance, where ACC_{rj}, ACC_{incp}, ACC_{wt}, ACC_{cp}, and ACC_{bg} denote the recognition accuracies for REJECT, INCOMPLETE, WAIT, COMPLETE, and BARGE, respectively. E2E denotes the mean end-to-end processing latency from video input to completion of interaction event prediction. All E2E latencies are measured on an NVIDIA RTX 4090 GPU. As shown in Table 1, to account for class imbalance in the test set, we weight the class-specific accuracies by their actual sample counts, yielding an overall accuracy of 94.99% and an end-to-end latency of 433.46 ms.

4.2. RGB–Pose Ablation

To evaluate the contribution of dual visual modeling to interaction event prediction, we examine RGB-only, Pose-only, and joint RGB+Pose inputs. We conduct three experiments: RGB-only retains only the RGB branch, Pose-only retains only the pose branch, and RGB+Pose uses the complete dual visual input.

As shown in Table 3, joint RGB and pose input outperforms either visual branch alone across all five event classes. Pose captures changes in hand motion through keypoint trajectories, while RGB provides complementary fine-grained information such as facial expressions and local appearance. Combining the two enables more accurate discrimination between brief pauses and ongoing signing, improving WAIT recognition. It also better distinguishes invalid input from new valid utterances, improving recognition of both REJECT and BARGE.

4.3. Downstream Task Evaluation

As an independent interaction controller, SignDuplex identifies turn states in continuous signing in real time and determines when to submit the current segment. To evaluate its plug-and-play capability, we select sign language translation as the downstream task and evaluate on the original sentence-level videos in How2Sign and continuous videos formed by concatenating multiple sentences. Given that existing sign language translation models are prone to performance degradation on long continuous input, we introduce Original (Reference), which segments and submits input using the ground-truth sentence boundaries annotated in the dataset, as a reference under ideal boundary conditions.

Table 4. Performance comparison of submission strategies on downstream sign language translation. Results are reported as single-sentence / continuous multi-sentence input.

Method	BLEU-4 ↑	T_{first} (s) ↓
Offline Translation	14.50 / 8.20	5.80 / 15.11
SL Segmentation [32]	13.70 / 11.85	6.12 / 6.48
Online CSLR/SLT [33]	12.80 / 10.90	5.98 / 6.26
SignDuplex	14.15 / 13.72	5.86 / 5.99
Original (Reference)	14.50 / 14.38	5.80 / 5.91

We consider three baselines: the offline strategy submits the complete video after it ends; the segmentation method of Moryossef et al. [32] is retrained on How2Sign and submits segments according to predicted boundaries; and the online method of Zuo et al. [33] is adapted to How2Sign and determines submission positions from online predictions. All methods use the same Uni-Sign model for translation. SignDuplex submits the current segment when COMPLETE is detected and continues processing the subsequent video. BLEU-4 measures translation quality [19]. The latency T_{first} denotes the time from the arrival of the first video frame until the system first submits a valid video segment to the downstream translation model, measuring system responsiveness. Table 4 presents the downstream sign language translation results for the different submission strategies.

The results show that SignDuplex improves translation of continuous multi-sentence input without modifying the downstream translation model. Mean first submission time decreases from 15.11 s for offline translation to 5.99 s, improving responsiveness while maintaining comparable sentence-level translation performance. These results validate the plug-and-play capability of SignDuplex as an independent interaction control layer: its predictions can directly drive downstream systems without changing the original task models. It can further extend to other sign language understanding, generation, and execution tasks, expanding the capabilities of existing sign language human–computer interaction systems.

5. CONCLUSION

We introduce SignDuplex, the first full-duplex interaction control framework for continuous signing. We define full-duplex sign language interaction events and combine RGB and pose information for real-time prediction. To address insufficient interaction data, we propose an automatic data construction pipeline and use it to construct the SignTurn dataset. As a plug-and-play interaction control module, SignDuplex improves translation quality and responsiveness for continuous signing without retraining the original task model, providing a feasible way to extend existing sign language systems with full-duplex interaction at low cost.

6. REFERENCES

[1] J. A. Hochgesang, "Motivated Look at Indicating Verbs in ASL (MoLo) Dataset," OSF, 2024, https://doi.org/10.17605/OSF.IO/VJP6W.

[2] J. A. Hochgesang and E. Shaw, "O5S5 ASL—Documenting the Experiences of the ASL Communities in the Time of COVID-19," OSF, 2024, https://osf.io/37w6t/.

[3] J. A. Hochgesang, K. Balzarini, A. Ivanov, K. Lipiz, S. J. Saenz, and G. Shi, "LING BING: Documenting Experiences in the Gallaudet Linguistics Department," OSF, 2025, https://doi.org/10.17605/OSF.IO/QEJND.

[4] A. Duarte, S. Palaskar, L. Ventura, D. Ghadiyaram, K. DeHaan, F. Metze, J. Torres, and X. Giro-i-Nieto, "How2Sign: A Large-Scale Multimodal Dataset for Continuous American Sign Language," in *Proc. IEEE/CVF CVPR*, 2021, pp. 2735–2744.

[5] R. Wong, N. C. Camgoz, and R. Bowden, "Sign2GPT: Leveraging Large Language Models for Gloss-Free Sign Language Translation," in *Proc. ICLR*, 2024.

[6] Z. Chen, B. Zhou, J. Li, J. Wan, Z. Lei, N. Jiang, Q. Lu, and G. Zhao, "Factorized Learning Assisted with Large Language Model for Gloss-free Sign Language Translation," in *Proc. LREC-COLING*, 2024, pp. 7071–7081.

[7] A. Yin, Z. Zhao, J. Liu, W. Jin, M. Zhang, X. Zeng, and X. He, "SimulSLT: End-to-End Simultaneous Sign Language Translation," in *Proc. ACM Multimedia*, 2021, pp. 4118–4127.

[8] C. Han, W. Ma, X. Tang, K. Hu, S. Wen, and Y. Xiang, "Boost-SLT: Boosting Sign Language Translation via a Plug-and-Play Diffusion-Based Semantic Enhancer," in *Proc. IEEE/CVF CVPR*, 2026, pp. 28828–28837.

[9] C. de Vos, F. Torreira, and S. C. Levinson, "Turn-Timing in Signed Conversations: Coordinating Stroke-to-Stroke Turn Boundaries," *Frontiers in Psychology*, vol. 6, pp. 268, 2015.

[10] S. Groeber and E. Pochon-Berger, "Turns and Turn-Taking in Sign Language Interaction: A Study of Turn-Final Holds," *Journal of Pragmatics*, vol. 65, pp. 121–136, 2014.

[11] G. Li, C. Wang, H. Xue, et al., "Easy Turn: Integrating Acoustic and Linguistic Modalities for Robust Turn-Taking in Full-Duplex Spoken Dialogue Systems," in *Proc. IEEE ICASSP*, 2026.

[12] J. Chen, Y. Hu, J. Li, et al., "FireRedChat: A Pluggable, Full-Duplex Voice Interaction System with Cascaded and Semi-Cascaded Implementations," *arXiv preprint arXiv:2509.06502*, 2025.

[13] A. Défossez, L. Mazaré, M. Orsini, A. Royer, P. Pérez, H. Jégou, E. Grave, and N. Zeghidour, "Moshi: a speech-text foundation model for real-time dialogue," *arXiv preprint arXiv:2410.00037*, 2024.

[14] Q. Zhang, L. Cheng, C. Deng, et al., "OmniFlatten: An End-to-end GPT Model for Seamless Voice Conversation," in *Proc. ACL*, 2025, pp. 14570–14580.

[15] J. Cui, B. Xu, C. Wang, et al., "MiniCPM-o 4.5: Towards Real-Time Full-Duplex Omni-Modal Interaction," *arXiv preprint arXiv:2604.27393*, 2026.

[16] Z. Liu, X. Zhang, Y. Li, et al., "Hierarchical Acoustic-Semantic Modeling: Modality Separation and Semantic Coherence for Full-Duplex SLMs," in *Proc. ACL*, 2026, pp. 9264–9280.

[17] Y. Chen, R. Zuo, F. Wei, Y. Wu, S. Liu, and B. Mak, "Two-Stream Network for Sign Language Recognition and Translation," in *Advances in Neural Information Processing Systems*, 2022, pp. 17043–17056.

[18] R. Zhang, R. Zhao, Z. Wu, L. Zhang, H. Zhang, and Y. Chen, "Dynamic Feature Fusion for Sign Language Translation Using HyperNetworks," in *Findings of NAACL*, 2025, pp. 6242–6254.

[19] Z. Li, W. Zhou, W. Zhao, K. Wu, H. Hu, and H. Li, "Uni-Sign: Toward Unified Sign Language Understanding at Scale," in *Proc. ICLR*, 2025.

[20] X. Zhai, B. Mustafa, A. Kolesnikov, and L. Beyer, "Sigmoid Loss for Language Image Pre-Training," in *Proc. IEEE/CVF ICCV*, 2023, pp. 11975–11986.

[21] T. Jiang, P. Lu, L. Zhang, N. Ma, R. Han, C. Lyu, Y. Li, and K. Chen, "RTMPose: Real-Time Multi-Person Pose Estimation based on MMPose," *arXiv preprint arXiv:2303.07399*, 2023.

[22] S. Yan, Y. Xiong, and D. Lin, "Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition," in *Proc. AAAI*, 2018, vol. 32, pp. 7444–7452.

[23] A. Yang et al., "Qwen3 Technical Report," *arXiv preprint arXiv:2505.09388*, 2025.

[24] S. Kita, I. van Gijn, and H. van der Hulst, "Movement Phases in Signs and Co-Speech Gestures, and Their Transcription by Human Coders," in *Gesture and Sign Language in Human-Computer Interaction*, vol. 1371 of *Lecture Notes in Computer Science*, pp. 23–35. Springer, 1998.

[25] I. Koulierakis, G. Siolas, E. Efthimiou, S.-E. Fotinea, and A.-G. Stafylopatis, "Recognition of Static Features in Sign Language Using Key-Points," in *Proc. LREC Workshop on the Representation and Processing of Sign Languages*, 2020, pp. 123–126.

[26] A. Moryossef, I. Tsochantaridis, R. Aharoni, S. Ebling, and S. Narayanan, "Real-Time Sign Language Detection using Human Pose Estimation," in *Proc. ECCV Workshops*, 2020, vol. 12536, pp. 237–248.

[27] N. Ravi, V. Gabeur, Y.-T. Hu, et al., "SAM 2: Segment Anything in Images and Videos," in *Proc. ICLR*, 2025.

[28] DeepSeek-AI, "DeepSeek-V3 Technical Report," *arXiv preprint arXiv:2412.19437*, 2024.

[29] H. Jiang, D. Sun, V. Jampani, M.-H. Yang, E. Learned-Miller, and J. Kautz, "Super SloMo: High Quality Estimation of Multiple Intermediate Frames for Video Interpolation," in *Proc. IEEE/CVF CVPR*, 2018, pp. 9000–9008.

[30] Z. Huang, T. Zhang, W. Heng, B. Shi, and S. Zhou, "Real-Time Intermediate Flow Estimation for Video Frame Interpolation," in *Proc. ECCV*, 2022, pp. 624–642.

[31] Team Wan, "Wan: Open and Advanced Large-Scale Video Generative Models," *arXiv preprint arXiv:2503.20314*, 2025.

[32] A. Moryossef, Z. Jiang, M. Müller, S. Ebling, and Y. Goldberg, "Linguistically Motivated Sign Language Segmentation," in *Findings of EMNLP*, 2023, pp. 12703–12724.

[33] R. Zuo, F. Wei, and B. Mak, "Towards Online Continuous Sign Language Recognition and Translation," in *Proc. EMNLP*, 2024, pp. 11050–11067.